

Resolution Semantics for Partial Algebras

Intrinsic Finite-Complement Separation and Observational Completion,

with a Lean-Checked Many-Sorted Encoding Bridge

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Abstract

We study one-sorted binary partial algebras through compatible total extensions that preserve the partial base pointwise while allowing only finitely many new states outside it. The underlying normalized Answer algebra is a concrete free compatible completion; that substrate is classical in character and is not the novelty claim of the paper. The main results concern the relative observation system built over it.

The first structural principle is *relative finite-pattern realization*: for every finite family of generated Answers there is one compatible finite-complement observer that is simultaneously injective on that family. Pairwise finite-tag separation is the two-point consequence, with the explicit constructor-size bound from the original separator. These observers induce a separated filtration and hence a Cauchy completion.

The second structural principle isolates the source of new completion points. If, at each observation stage, a generated unary family is interpreted as a deterministic trajectory admitting a uniformly finite trajectory code, then factorial sampling is Cauchy. This finite-dynamics step is standard and is not claimed as new. The Resolution-specific anti-limit mechanism is complementary: for a syntactically growing unary orbit, a candidate-tailored finite-pattern observer realizes the candidate and a long orbit prefix exactly, sends the next unseen iterate to an absorbing overflow state, and therefore separates every sufficiently late iterate from the candidate. Thus trajectory compression plus finite-pattern escape forces the observational completion to be proper.

Both the exact finite-base properness criterion and the arbitrary-base base-fixing-context theorem factor through this combined mechanism. In the old-fixing case the stage- n indistinguishable factorial pair sharpens to $c_{(n+1)!} \equiv_n c_{(n+2)!}$, giving unbounded separation ranks. Natural and integer arithmetic instantiate the arbitrary-base criterion while retaining zero-denominator applications as structured Answers rather than ordinary numbers. The completed algebra preserves exactly the universal equations in the language with named base constants.

Finally, a separate Lean-checked many-sorted development formalizes the operations-only fragment of Diaconescu's $\Gamma\text{-}\alpha\text{-}\beta\text{-}\gamma$ encoding for arbitrary finite arities, including the satisfaction condition, adjunction laws, semantic consequence, and the fixed-signature initiality results used here. Its singleton-sort, no-*TF*, binary-*PF* specialization is canonically equivalent to the binary Resolution implementation. Lean 4.33.0 checks the complete formal package.

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1 Introduction

Partial algebras distinguish defined applications from undefined ones. A standard way to totalize such a structure is to pass to a free compatible completion, retaining enough syntax to represent unresolved applications. In the one-sorted binary setting used here, an expression is resolved bottom-up: applications already defined in the base algebra reduce to embedded base values, while unresolved applications remain finite suspended trees. The resulting generated Answer algebra is a concrete presentation of a free compatible extension. Free completion itself is established theory, and we do not claim novelty for that substrate; see the Schmidt-construction literature and the many-sorted partial-to-total encoding of Diaconescu [12, 16].

The paper studies a more specific relative observation problem. Let D be a binary partial algebra with carrier A . We consider compatible total extensions whose embedding of A is injective and whose complement over that pointwise-preserved copy of A is finite. Crucially, A itself need not be finite. Thus the observation budget measures only genuinely new semantic states beyond the original partial base.

This relative observer class gives rise to two structural mechanisms.

Finite-pattern realization. For every finite list of generated Answers there is one compatible finite-complement observer that is simultaneously injective on the whole list. The observer retains the full base pointwise, assigns finite tags to the selected suspended subterms, and sends unselected behavior to one overflow state. Pairwise separation is therefore not the primitive phenomenon: it is the two-point instance of a finite-pattern theorem. This is the first master principle of the paper.

Compression and escape. The finite observers induce a separated filtration and a Cauchy completion. To understand when that completion adds new points, we separate two independent mechanisms. First, if a chosen generated unary family is interpreted at every finite stage as a deterministic trajectory with uniformly finite trajectory complexity, then the factorially sampled family is Cauchy. This uses only pigeonhole periodicity and factorial divisibility; it is standard finite-state dynamics and is not claimed as new. Second, a syntactically growing unary orbit has a Resolution-specific anti-limit property. Given a proposed finite Answer x , the finite-pattern construction can retain x and a sufficiently long orbit prefix exactly. The next unseen iterate cannot belong to the selected finite pattern, so it is sent to overflow; overflow is absorbing along the continuation. Hence every sufficiently late iterate is separated from x . This is the finite-pattern escape mechanism.

The combination is the conceptual center of the paper:

finite trajectory compression + finite-pattern escape $\implies$ proper observational completion.

The two halves play opposite roles. Compression makes the factorial sample indistinguishable at every fixed finite stage, while escape prevents it from converging to any finite generated Answer. Their interaction forces a new completion point.

This organization unifies results that can otherwise look unrelated. When the base is finitely coded, the whole stage observer is finite, so trajectory compression is automatic and one obtains an exact criterion: the completion is proper exactly when the base evaluator is not total. For an arbitrary, possibly infinite base, a unary context that fixes every base

element pointwise still compresses the relevant orbit even though the ambient observer carrier may be infinite. The same finite-pattern escape argument supplies nonconvergence. Thus finite-base and old-fixing properness are instances of one combined mechanism, not two independent completion tricks.

The old-fixing analysis also yields quantitative information. At stage n there are only $n + 1$ external states—the n tags plus overflow—unless the orbit enters the pointwise-fixed base, after which it stabilizes. Consequently

$$c_{(n+1)!} \equiv_n c_{(n+2)!},$$

for a distinct pair of iterates, and the least finite-tag separation ranks are unbounded. The factorial indices are only a synchronization device for finite periods; the mathematical content lies in the relative observer structure that makes such finite compression coexist with finite-pattern escape.

The completion also preserves exactly the universal equations in the language with named base constants. Natural and integer arithmetic provide concrete infinite-base examples: division by zero remains a structured unresolved Answer, while the context $x \mapsto x + 0$ produces the compressed escaping orbit that witnesses a proper completion.

To locate the construction precisely within established partial-algebra theory, we additionally give a Lean-checked many-sorted implementation of the operations-only fragment of Diaconescu’s encoding. It supports arbitrary finite arities and separate total and partial operation families, proves the $\Gamma\text{-}\alpha\text{-}\beta\text{-}\gamma$ satisfaction and adjunction results used here, and canonically identifies its singleton-sort binary specialization with the original Resolution implementation. This bridge is supporting provenance, not a third novelty claim.

The formal development is checked in Lean 4.33.0. The publication-facing master statements are `ResolutionSemantics.MasterTheorems.relativeFinitePatternRealization`, `ResolutionSemantics.MasterTheorems.orbitCompressionCauchy`, `ResolutionSemantics.MasterTheorems.finitePatternEscapeNoGeneratedLimit`, and `ResolutionSemantics.MasterTheorems.compressedEscapingOrbitEmbeddingNotSurjective`.

1.1 Main results and paper organization

The logical dependency of the paper is as follows.

1. [Section 2](#) records the normalized free compatible completion used as the substrate.
2. [Section 3](#) develops intrinsic finite-complement observers and proves relative finite-pattern realization.
3. [Section 4](#) constructs the observational filtration and isolates trajectory compression, finite-pattern escape, and their combined properness theorem. The finite-base and old-fixing criteria follow as instances.
4. [Sections 5 and 6](#) give exact equational conservativity and arithmetic examples.
5. [Section 7](#) verifies the many-sorted encoding bridge.
6. [Sections 8 and 10](#) delimit the contribution and remaining open problems.

The paper deliberately distinguishes standard machinery from the Resolution-specific statements. Free compatible completion, ordinary finite-state periodicity, factorial synchronization, and generic Cauchy completion are background mechanisms. The contribution claimed here is the relative observer framework over a pointwise-preserved partial base,

its finite-pattern realization theorem, and the way finite-pattern escape interacts with trajectory-level compression to force proper observational completions, including over infinite bases.

2 Generated Answers as a Free Compatible Completion

This section records the algebraic substrate used by the observational results. Its role is foundational rather than novel: the normalized construction is a concrete binary presentation of a free compatible completion.

Let D have carrier A , binary operation symbols Σ , and evaluator

$$\text{eval}_D : \Sigma \rightarrow A \rightarrow A \rightarrow \text{Option}(A).$$

Expressions and raw Answers are generated by

$$e ::= \text{val}(a) \mid \text{app}(f, e_1, e_2), \quad r ::= \text{old}(a) \mid \text{susp}(f, r_1, r_2).$$

Define the one-step resolution operation by

$$\text{liftOp}_D(f, x, y) = \begin{cases} \text{old}(c), & x = \text{old}(a), y = \text{old}(b), \text{eval}_D(f, a, b) = \text{some}(c), \\ \text{susp}(f, x, y), & \text{otherwise.} \end{cases}$$

The evaluator Res_D recursively resolves both children and then applies liftOp_D . The generated Answer algebra $R(D)$ is the image of Res_D , with base embedding $\eta_0(a) = \text{old}(a)$ and operations induced by liftOp_D .

Proposition 2.1 (Conservativity on the base). *If $\text{eval}_D(f, a, b) = \text{some}(c)$, then*

$$f_{R(D)}(\eta_0(a), \eta_0(b)) = \eta_0(c).$$

The base embedding η_0 is injective.

A compatible total algebra over D consists of a total Σ -algebra T and a map $i : A \rightarrow T$ satisfying

$$\text{eval}_D(f, a, b) = \text{some}(c) \implies f_T(i(a), i(b)) = i(c).$$

The map need not be injective for the free universal property.

Every raw Answer has a canonical fold into T :

$$\Phi_T(\text{old}(a)) = i(a), \quad \Phi_T(\text{susp}(f, x, y)) = f_T(\Phi_T(x), \Phi_T(y)).$$

Compatibility is exactly what is needed to show that this fold respects the resolution step.

Theorem 2.2 (Free compatible completion). *For every compatible total algebra T over D , there is a unique homomorphism*

$$\Phi_T : R(D) \rightarrow T$$

with $\Phi_T \circ \eta_0 = i$.

The checked publication-facing declaration is `ResolutionSemantics.freeCompletionUniversal`. The expression algebra modulo equality after resolution is likewise canonically equivalent to the generated Answer algebra; the two directions are exposed by `ResolutionSemantics.expressionKernelQuotientInjective` and `ResolutionSemantics.expressionKernelQuotientSurjective`.

Remark 2.3 (Positioning). The theorem above should be read as a concrete normalized presentation of an established free-completion phenomenon, not as the main novelty claim. Its purpose in this paper is to provide canonical generated syntax and canonical interpretation maps into compatible observers. All later separation and completion statements are relative to those interpretation maps.

Generated Answers are normal forms: a suspended node cannot have two old children on which the base evaluator is defined, because such a node would have reduced. This simple normalization invariant is the key local fact used by the finite-pattern observers in the next section.

3 Relative Finite-Pattern Realization

The free-completion theorem gives canonical maps into compatible total extensions, but it does not say which distinctions between suspended Answers are visible in independently specified models. We now restrict attention to observers that preserve the base pointwise and have only finitely many states outside it.

Let T be a compatible total extension with injective base embedding $i : A \hookrightarrow T$. Its genuine outside part is

$$\text{Out}_D(T) = \{z \in T \mid \forall a \in A, z \neq i(a)\}.$$

An intrinsic observer has budget n when $\text{Out}_D(T)$ injects into $\text{Fin}(n+1)$. The base A may be infinite. The budget counts only states beyond the fixed copy of A .

A canonical stage- n representative has carrier

$$A \sqcup \text{Fin}(n) \sqcup \{\star\},$$

with total operations that preserve every application defined in D . The formal representation theorem identifies separation by arbitrary intrinsic budget- n observers with separation by these finite-tag models. Write $x \equiv_n y$ when every stage- n finite-tag model interprets x and y equally.

The original pairwise separator can be strengthened. Instead of selecting the subterms of two roots, select the union of the finite subterm closures of an arbitrary finite family. Old leaves keep their base values. Every selected suspended node receives the tag of its first occurrence in the selected list, and all other behavior is sent to overflow. A finite lookup table reproduces each selected normal suspension exactly. Induction over the selected closure then shows that every selected generated Answer evaluates to its injective encoding.

Theorem 3.1 (Relative finite-pattern realization). *Let $x_1, \dots, x_k \in R(D)$ be any finite family. There exist n and one compatible stage- n finite-tag observer T such that the canonical interpretation*

$$\Phi_T : R(D) \rightarrow T$$

is injective on $\{x_1, \dots, x_k\}$.

Lean checks the theorem as `ResolutionSemantics.MasterTheorems.relativeFinitePatternRealization`. The construction fixes the entire base carrier pointwise; only the selected suspended pattern consumes finite tags.

Corollary 3.2 (Intrinsic finite-complement separation). *For every distinct $x, y \in R(D)$ there is a compatible total extension whose outside part has at most $|x| + |y| + 1$ states and in which x and y have different interpretations. Equivalently,*

$$(\forall n, x \equiv_n y) \iff x = y.$$

The quantitative pair bound remains exposed by `ResolutionSemantics.finiteSeparationBound`; the intrinsic observer-class statement is `ResolutionSemantics.MasterTheorems.relativeFiniteComplementSeparation`. The finite-pattern theorem is structurally stronger than the pairwise bound, while the pairwise implementation retains the sharper explicit estimate $n = |x| + |y|$.

Remark 3.3 (What is and is not being claimed). The selected-subterm plus sink construction has standard automata-theoretic antecedents [8]. The contribution claimed here is not the existence of a sink automaton in isolation. It is the relative realization statement for observers that must preserve a possibly infinite partial base pointwise while budgeting only the complement. In particular, the codomain need not be finite when A is infinite.

If the base admits an injection into $\text{Fin}(m)$, then the same observer becomes an ordinary finite algebra with total-state bound

$$m + |x| + |y| + 1.$$

Hence $R(D)$ is residually finite in the ordinary sense for finitely coded bases. This qualitative finite-base consequence is classical in the finite-signature ground-equational setting; the relative construction is used because it remains meaningful when the base itself is infinite.

Proposition 3.4 (Closure of the base image). *The image of A in $R(D)$ is closed under every generated total operation if and only if the evaluator of D is total.*

Thus in a genuinely partial example the preserved base is not a total subalgebra of $R(D)$. This is one reason ordinary finite-Rees-index language does not literally describe the relative observer situation.

4 Observational Filtration, Compression, and Escape

The stage relations $\equiv_n$ form a descending filtration: agreement at a larger finite-tag budget implies agreement at every smaller budget. By [corollary 3.2](#), their intersection is equality, so the filtered space is separated. A sequence is Cauchy when, for each fixed n , its tail is pairwise $\equiv_n$ -equivalent. Quotienting Cauchy sequences by eventual agreement at every stage gives the observational completion $\widehat{R(D)}$. The constant-sequence map is injective and stage-dense, and standard diagonalization gives completeness.

The central question is when this completion contains genuinely new points. The proof is best understood by separating two logically independent mechanisms.

4.1 Master II: trajectory compression

Fix a generated unary family $t_0, t_1, \dots$. In a stage- n observer, suppose its interpretation is the orbit of one deterministic self-map

$$s_{k+1} = F_n(s_k).$$

We do not require the entire observer carrier to be finite. It is enough that the single trajectory (s_k) admit a finite code with bound $N(n)$ such that, for ordered times $i \leq j$, equality of codes implies $s_i = s_j$.

Theorem 4.1 (Finite trajectory compression implies factorial Cauchy). *If such a finite trajectory code exists uniformly at every stage, then the factorially sampled generated sequence*

$$t_0!, t_1!, t_2!, \dots$$

is Cauchy for the finite-tag filtration.

The formal statement is:

`ResolutionSemantics.MasterTheorems.orbitCompressionCauchy.`

The proof is standard finite dynamics. Among the first $N(n) + 1$ coded orbit positions some state repeats. The resulting eventual period is at most $N(n)$ and therefore divides $N(n)!$. Sufficiently late factorial indices are multiples of this period, so all late factorial samples agree at stage n .

The distinction between trajectory finiteness and carrier finiteness is essential. For finite bases the whole stage observer can be coded finitely. In the infinite-base old-fixing case the observer carrier remains infinite, but the relevant orbit admits a finite quotient after collapsing the pointwise fixed old states to a settled class. Thus both cases instantiate the same dynamic theorem.

4.2 Finite-pattern escape

Compression explains Cauchy behavior but cannot by itself imply that the completion is proper: a Cauchy sequence might converge to a generated Answer. The anti-limit argument comes from the concrete finite-pattern realization mechanism.

Assume the generated family has a growing unary syntactic form

$$t_{k+1} = \text{susp}(u, t_k, \text{old}(e))$$

for a fixed operation symbol u and fixed right base argument e . Every step adds a new suspended constructor, so constructor size strictly increases.

Fix a candidate generated Answer x . Choose an orbit prefix t_N longer than x and build the finite-pattern observer from the two roots x and t_N . The observer evaluates every selected subterm to its exact injective encoding. The next orbit node t_{N+1} is too large to be a subterm of either selected root. Hence the transition from the encoded prefix to the fixed right argument misses the selected lookup table and is sent to $\star$. No selected node has overflow as an encoded left child, so the same unary context keeps overflow fixed thereafter.

Theorem 4.2 (Finite-pattern escape). *For every candidate $x \in R(D)$, a candidate-dependent finite-tag observer separates x from every sufficiently late term of the factorially sampled growing unary orbit. Consequently the factorial sample has no generated limit.*

Lean checks this as:

`ResolutionSemantics.MasterTheorems.finitePatternEscapeNoGeneratedLimit.`

Unlike [theorem 4.1](#), this proof does not use finite-state periodicity. Its input is the finite-pattern observer together with syntactic growth and absorbing overflow.

4.3 Combined properness theorem

Theorem 4.3 (Compression–escape properness). *Suppose a generated unary family satisfies both:*

1. *at each finite observation stage its interpreted trajectory admits a uniformly finite trajectory code; and*
2. *its generated syntax grows by repeated application of a fixed unary context of the form $t \mapsto \text{susp}(u, t, \text{old}(e))$.*

Then the factorial sample is Cauchy and has no generated limit. Therefore the generated filtered space is incomplete and the canonical embedding

$$R(D) \longrightarrow \widehat{R(D)}$$

is not surjective.

The publication-facing Lean theorem is:

`ResolutionSemantics.MasterTheorems.compressedEscapingOrbitEmbeddingNotSurjective.`

The incompleteness form is:

`ResolutionSemantics.MasterTheorems.compressedEscapingOrbitNotComplete.`

This is the structural center of the properness theory. Compression and escape are complementary rather than redundant: the first makes late terms observationally indistinguishable at each fixed stage, while the second shows that no finite generated Answer can absorb that Cauchy behavior as its limit.

4.4 Finite-base instance

Suppose A injects into $\text{Fin}(m)$ and some base application $\text{eval}_D(f, a, z)$ is undefined. Let $c_0 = \text{old}(a)$ and iterate the unresolved left comb $c_{k+1} = f(c_k, \text{old}(z))$. A stage- n observer has at most $m + n + 1$ coded states, so its comb trajectory is finitely coded. The comb also has the growing unary syntax required by finite-pattern escape.

Theorem 4.4 (Finite-base properness criterion). *For a finitely coded base, the canonical completion embedding is surjective if and only if the evaluator of D is total. Equivalently, the completion is proper exactly when some base application is undefined.*

This is exposed through:

`ResolutionSemantics.MasterTheorems.finiteBasePropernessCriterion.`

The reverse direction is elementary: if the evaluator is total, every generated Answer reduces to an embedded base value and every Cauchy sequence is eventually constant at a separating finite stage.

4.5 Infinite-base old-fixing instance

No finite code of the whole base is needed when the orbit is constrained by a context that fixes old elements pointwise. Suppose

$$\text{eval}_D(g, a, b) = \text{none}, \quad \text{eval}_D(u, x, e) = \text{some}(x) \quad (x \in A),$$

and define

$$c_0 = \text{susp}(g, \text{old}(a), \text{old}(b)), \quad c_{k+1} = \text{susp}(u, c_k, \text{old}(e)).$$

In any stage- n observer, if the orbit enters the old base then the old-fixing law makes it constant. Otherwise it stays entirely in the n tags plus $\star$, hence in only $n + 1$ external states. This yields the sharp factorial synchronization

$$c_{(n+1)!} \equiv_n c_{(n+2)!}.$$

Theorem 4.5 (Old-fixing properness and unbounded rank). *Under the hypotheses above, the observational completion is proper. Moreover, for every n the pair $c_{(n+1)!}, c_{(n+2)!}$ is distinct but agrees at stage n , so its least finite-tag separation rank is greater than n .*

The combined-master factorization is:

`ResolutionSemantics.MasterTheorems.oldFixingPropernessViaCombinedMaster.`

The sharpened rank theorem is:

`ResolutionSemantics.MasterTheorems.oldFixingRanksUnbounded.`

The finite-base and old-fixing hypotheses are incomparable. A one-point partial algebra can be finitely coded while admitting no base-fixing unary context; natural arithmetic has the context $x \mapsto x + 0$ but no finite code of its carrier. The common theorem is not a common hypothesis on the base, but a common mechanism on the relevant observer trajectory.

4.6 Completed operations and universal extension

Every generated primitive operation preserves every stage relation and extends pointwise to the Cauchy completion. The completed algebra is a compatible total extension with injective base embedding. Relative to complete separated filtered compatible targets with jointly nonexpansive primitive operations, the completion satisfies the expected extension property. Lean exposes completeness as:

`ResolutionSemantics.completionComplete.`

The relative initiality package is:

`Resolution.External.FilteredTotalAlg.completedResolutionFilteredAlgebra_initial.`

The observational filtration differs from prefix-depth observation on trees. An unresolved comb can be Cauchy for the classical prefix-depth metric while a single finite-tag observer separates adjacent combs at every depth. The formal contrast is:

`Resolution.Probe.depthCauchy_not_observationalCauchy.`

Thus the completion studied here is not merely the classical infinite-tree completion under another notation.

5 Exact Equational Conservativity

The completion adds observational limit points, but it does not create new universal equations in the language containing variables, named base constants, and the primitive binary operations.

Term evaluation commutes with the canonical embedding $R(D) \rightarrow \widehat{R(D)}$. Moreover, every term operation preserves each stage relation because the primitive operations do. Given a completed assignment and a finite stage n , stage-density supplies generated approximants for all variables. If an equation holds for all generated assignments, the two completed term values are therefore equivalent at every finite stage. Separatedness upgrades agreement at all stages to equality. The converse follows immediately by restricting a completed equation to embedded generated assignments.

Theorem 5.1 (Exact equation preservation). *For every constant-bearing algebraic equation $s = t$,*

$$R(D) \models s = t \iff \widehat{R(D)} \models s = t.$$

The public Lean theorem is `ResolutionSemantics.equationConservative.`

Remark 5.2 (Scope). This is a theorem about universal equations. Arbitrary quasi-equations do not follow from density alone; conditional laws require additional hypotheses on the premises. No unrestricted quasi-equational conservativity claim is made.

6 Arithmetic Examples

The construction is not tied to arithmetic, but natural and integer examples make the arbitrary-base properness mechanism concrete. In both cases division is partial at denominator zero, while addition supplies the base-fixing context $x \mapsto x + 0$.

6.1 Natural numbers

Take $+$, $\cdot$, and $/$ on $\mathbb{N}$, with division undefined exactly at zero denominator. For $b \neq 0$, resolution returns the embedded natural a/b . At zero,

$$a/0 = \text{susp}(\text{div}, \text{old}(a), \text{old}(0)).$$

Distinct numerators give distinct singular Answers, and no singular Answer is an embedded natural. Lean exposes these statements as `ResolutionSemantics.NatDivision.singularFamilyInjective` and `ResolutionSemantics.NatDivision.singularFamilyDisjoint`.

The base carrier is infinite and admits no finite injection into $\text{Fin}(m)$. Nevertheless, addition satisfies $a + 0 = a$ for every base natural. Starting from the undefined seed $0/0$ and iterating the context $x \mapsto x + 0$ gives the old-fixing orbit of [theorem 4.5](#). Hence its factorial sample is Cauchy with no generated limit, the observational completion is proper, and finite-tag separation ranks are unbounded. The criterion is exposed by `ResolutionSemantics.NatDivision.oldFixingCriterion` and the rank consequence by `ResolutionSemantics.NatDivision.separationRanksUnbounded`.

This example also illustrates why the filtration is not merely syntactic depth. A different iterated-division comb can be separated already at stage zero by using unused base naturals as semantic memory, while the repeated- $+0$ orbit cannot: preservation forces every base natural to be fixed by the context. Thus the relative observer class can see deep syntax aggressively in one family and compress it strongly in another.

6.2 Integers

The integer instance additionally includes subtraction. Zero-denominator applications again remain structured suspended Answers rather than ordinary integers. The singular family is injective and completed $0/0$ remains outside the embedded integer image; see `ResolutionSemantics.IntDivision.singularFamilyInjective` and `ResolutionSemantics.IntDivision.zeroDivZeroNotOld`.

Because $a + 0 = a$ also holds over $\mathbb{Z}$, the same compression–escape mechanism proves properness and unbounded separation rank. The relevant public declarations include `ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective` and `ResolutionSemantics.IntDivision.separationRanksUnbounded`.

Resolution does not assign an ordinary numerical value to $0/0$. It preserves its unresolved provenance as a structured Answer in a conservative total extension.

This differs from meadows, common meadows, and wheels, where singular expressions are totalized by algebraic conventions or designated exceptional values [2, 3, 6]. The goal here is not to replace those systems, but to retain the computation that failed and let compatible observers decide how much of that structure to remember.

7 A Lean-Checked Bridge to Diaconescu's Encoding

The main separation and completion theorems above are one-sorted and binary. Their free-completion substrate sits inside a broader partial-to-total theory. To make that provenance explicit, the repository contains an independent many-sorted Lean development of the operations-only fragment of Diaconescu's encoding [12].

A many-sorted signature has sorts S , designated-total operation symbols TF , and partial operation symbols PF , each with arbitrary finite arity and typed argument and result sorts. Encoded total models extend the data carriers with an auxiliary truth carrier and existence-equality operations. The Horn conditions Γ ensure that the defined elements form the partial reduct $\beta(M)$.

The translation α sends existence-equality atoms to equality with the truth constant and guards universal quantification by definedness. Lean proves the term soundness and completeness lemmas and hence the satisfaction condition

$$M \models \alpha(\rho) \iff \beta(M) \models \rho.$$

The relevant declarations are `Resolution.Diaconescu.ManySorted.term_evalPartial_sound`, `Resolution.Diaconescu.ManySorted.term_evalPartial_complete`, and `Resolution.Diaconescu.ManySorted.alpha_satisfaction_condition`.

For every partial algebra D , the canonical model $\gamma(D)$ uses normalized formal applications as its total carrier. Lean checks $\gamma(D) \models \Gamma$ and a canonical partial-algebra equivalence $\beta(\gamma(D)) \simeq D$ through `Resolution.Diaconescu.ManySorted.gamma_satisfiesGamma` and `Resolution.Diaconescu.ManySorted.betaGammaPartialAlgEquiv`. The universal lift from a partial homomorphism into a reduct is formalized as a hom-set equivalence; identity, composition, naturality, unit/counit behavior, and both triangle identities are checked.

At fixed signature and universe bound, the development proves the initiality transfers needed for the operations-only quasi-existence theories considered here. A conditional term-model construction supplies the initial translated encoded model, and the β reduct yields the corresponding initial partial model. Semantic consequence is preserved and reflected by the translation. Representative declarations include `Resolution.Diaconescu.ManySorted.gammaBetaHomEquiv`, `Resolution.Diaconescu.ManySorted.gamma_preserves_initiality`, `Resolution.Diaconescu.ManySorted.beta_of_initial_encoded_is_initial`, `Resolution.Diaconescu.ManySorted.semantic_consequence_equivalence`, and `Resolution.Diaconescu.InitialExistence.quasiTheory_has_initial_partial_model`.

Finally, when the signature has one sort, no designated-total symbols, and only binary partial symbols, Lean constructs canonical equivalences between the many-sorted generated carrier and the original binary Resolution carrier, and between the corresponding encoded models. The principal declarations are `Resolution.Diaconescu.BinarySpecialization.generatedEquiv` and `Resolution.Diaconescu.BinarySpecialization.gammaEquiv`.

Remark 7.1 (Role of the bridge). The bridge verifies that the binary implementation used by the observer theory is a specialization of an established many-sorted partial-to-total encoding. It is supporting formal provenance. We do not claim Diaconescu's encoding, the adjunction, or the existence of free partial completions as new results. Nor does the present paper extend the finite-complement separation and compression-escape completion theory to the full many-sorted setting; that remains an open direction.

8 Relation to Prior Work

8.1 Free completion and partial algebras

The free compatible completion underlying [section 2](#) belongs to an established line of work on partial algebras and Schmidt-type completions [16]. Diaconescu’s many-sorted encoding provides a closely related partial-to-total perspective [12]. These results motivate the substrate but are not novelty claims of the present paper.

8.2 Recognizable trees and finite separators

For finite bases and finite signatures, the normalized ground-equational presentation lies within classical tree-automata and recognizability theory [4, 8, 10, 13]. Finite selected subterm closures plus a sink state are familiar constructions. The relative feature studied here is the requirement that every observer preserve an arbitrary partial base pointwise while only the complement over that base is budgeted. This permits infinite codomains with finite external complexity.

8.3 Filtered and profinite completions

Cauchy completion of separated filtrations and factorial synchronization of finite deterministic dynamics are standard. Profinite completions organize observations by finite quotients or finite recognizing algebras [7, 11, 14]. Resolution stages differ because the preserved base may be infinite. The compression theorem deliberately isolates the standard dynamic ingredient: only the trajectory relevant to the chosen unary family needs a finite code. The new structural interaction is with the relative finite-pattern escape observer that prevents convergence to any finite generated candidate.

8.4 Finite complements and Rees-index analogies

Finite-complement extensions and residual finiteness are classical in semigroup theory [5, 15]. The analogy is useful but not literal here. In a genuinely partial Resolution algebra, the embedded old carrier is not closed under the generated total operations; by [proposition 3.4](#), closure occurs exactly when the evaluator was already total. Hence the observer class is better described as relative to a pointwise-fixed partial base than as an ordinary finite-Rees-index extension.

8.5 Tree metrics

Classical infinite-tree completions are based on prefix depth [1, 9]. Resolution uses semantic finite-tag observation instead. The formal comb counterexample in [section 4](#) shows that the two Cauchy notions differ: a sequence may be depth-Cauchy while a fixed low-budget Resolution observer continues to distinguish adjacent terms.

8.6 Contribution and scope

The contribution claimed by this paper is concentrated in the following package:

1. an intrinsic finite-complement observer class over a pointwise-fixed, possibly infinite partial base, together with finite-pattern realization;
2. the induced observational filtration and its trajectory-level compression theorem;
3. the Resolution-specific finite-pattern escape mechanism and the resulting compression-escape properness theorem;
4. the exact finite-base criterion, infinite-base old-fixing instance, sharper unbounded-rank witnesses, and exact equational conservativity.

The factorial-periodicity argument itself is treated as standard machinery. Likewise, the many-sorted Diaconescu development is a verified compatibility and provenance layer rather than an independent novelty claim.

9 Formalization and Trust Boundary

The repository pins Lean 4.33.0. The proof package contains a small publication-facing facade in `ResolutionMasterTheorems.lean` that exposes the structural results used in the reorganized paper. In particular, the following declarations are checked directly:

```
#check ResolutionSemantics.MasterTheorems.relativeFinitePatternRealization
#check ResolutionSemantics.MasterTheorems.orbitCompressionCauchy
#check ResolutionSemantics.MasterTheorems.finitePatternEscapeNoGeneratedLimit
#check ResolutionSemantics.MasterTheorems.compressedEscapingOrbitNotComplete
#check ResolutionSemantics.MasterTheorems.
  compressedEscapingOrbitEmbeddingNotSurjective
#check ResolutionSemantics.MasterTheorems.finiteBasePropernessCriterion
#check ResolutionSemantics.MasterTheorems.oldFixingPropernessViaCombinedMaster
#check ResolutionSemantics.MasterTheorems.oldFixingRanksUnbounded
```

The finite-pattern result is implemented in `ResolutionFinitePatternRealization.lean`. The trajectory-compression abstraction is in `ResolutionOrbitCompressionMaster.lean`; its finite code applies only to the relevant deterministic orbit and does not require a finite ambient observer carrier. The anti-limit theorem is implemented in `ResolutionFinitePatternAntiLimit.lean`; it formalizes the selected candidate/prefix construction, overflow entry, and overflow absorption. Both the finite-base and old-fixing properness theorems are re-derived as instances of the combined theorem.

The command

```
bash scripts/verify.sh
```

checks the manuscript-linked public API, compiles the proof dependency chain, rejects direct proof holes or local axioms in the paper sources, and runs the axiom audit. The accepted theorem dependencies are restricted to `propext`, `Classical.choice`, and `Quot.sound`.

The formalization uses classical finite indexing and witness selection in explicit places. The resulting trust claim is therefore relative to Lean’s kernel, the imported standard library, and the ordinary classical principles reported by the audit. No foundation-independent consistency claim is made.

10 Limitations and Open Questions

The structural reduction clarifies several boundaries rather than eliminating them.

1. The finite-pattern realization theorem is proved for the main one-sorted binary Resolution kernel. An independent Lean module checks the same realization mechanism for arbitrary finite arities on finite child-closed normalized raw patterns, but the full observational completion theory has not yet been lifted to the many-sorted setting.
2. The pairwise constructor-size tag bound is a general upper bound, not an optimal state-complexity theorem. Determining the optimal relative separation rank as a function of Answer size and signature/base hypotheses remains open.
3. The finite-base theorem gives an exact properness classification for finitely coded carriers. The old-fixing condition is a sufficient infinite-base criterion, not a necessary one. A full arbitrary-carrier characterization of properness remains open.
4. Trajectory compression is deliberately weak: it constrains only the observer orbit relevant to a selected generated family. Other trajectories in the same observer may have unbounded or genuinely infinite complexity. This is a feature of the theorem, but it limits conclusions about global finite-state structure.
5. Finite-pattern escape is currently formalized for a fixed right unary syntactic context $t \mapsto \text{susp}(u, t, \text{old}(e))$. More general one-hole contexts and arbitrary finitary contexts are natural extensions.
6. The many-sorted bridge checks the concrete operations-only fragment, fixed-signature adjunction laws, semantic consequence, and the initial-model construction used here. It does not formalize the full surrounding institution/comorphism infrastructure or arbitrary signature morphisms from Diaconescu’s broader theory.
7. Exact conservativity is proved for universal equations with named base constants. Arbitrary conditional equations require additional hypotheses.
8. The construction is semantic, not an efficiency claim. Exact provenance retention can produce large finite Answers and large candidate-tailored observer tables.

Open Question 10.1 (Relative compression–escape criteria). Can the combined properness theorem be characterized directly in terms of the partial algebra D , without first exhibiting a particular compressed escaping unary orbit? Such a criterion would extend the exact finite-base theorem beyond finitely coded carriers.

Open Question 10.2 (General contexts). How far does finite-pattern escape extend from the fixed-right unary context to arbitrary finite one-hole contexts, many-sorted signatures, or finitely branching context systems? The existing finite-pattern realization theorem suggests that the anti-limit mechanism is more general than the current unary formalization.

11 Conclusion

Resolution semantics starts from a familiar free-compatible-completion idea: defined base applications reduce, while undefined applications remain structured finite Answers. The present paper’s contribution lies in what is built over that substrate.

The first master principle is relative finite-pattern realization. A finite family of generated Answers can be realized simultaneously inside one compatible observer that fixes the entire partial base pointwise and uses only finitely many new states beyond it. Pairwise

finite-complement separation is the two-point consequence, and finite-tag models provide canonical stage representatives. These stages induce a separated observational filtration.

The second part of the story explains why its completion can be genuinely larger than the generated Answer algebra. Finite trajectory compression makes factorial samples Cauchy even when the ambient observer carrier is infinite. Finite-pattern escape supplies the opposite force: every finite generated candidate can be retained exactly while a sufficiently late continuation of a growing orbit is forced into absorbing overflow. Thus the same relative observer technology both compresses infinite behavior at every fixed stage and exposes escape beyond every finite candidate. Their interaction forces new completion points.

This compression–escape theorem unifies the finite-base and infinite-base old-fixing properness results. For finitely coded bases, properness is exactly equivalent to partiality of the evaluator. For arbitrary bases, a pointwise base-fixing unary context supplies a compressed trajectory; the sharpened stage- n witness $c_{(n+1)!} \equiv_n c_{(n+2)!}$ yields unbounded separation rank. Natural and integer arithmetic instantiate this mechanism through the context $x \mapsto x + 0$, while singular division remains structured rather than being assigned an ordinary numerical value.

The observational completion carries canonical extensions of the primitive operations and preserves exactly the universal equations of the generated algebra in the language with named base constants. Its filtration is not the classical prefix-depth filtration on trees: finite semantic observers can compress or distinguish arbitrarily deep syntax according to compatibility with the preserved partial base.

The many-sorted Lean development provides a separate provenance result. It checks the relevant $\Gamma\text{-}\alpha\text{-}\beta\text{-}\gamma$ encoding machinery and proves that the original binary Resolution core is the expected single-sorted specialization. This distinguishes inherited partial-to-total infrastructure from the relative observation and completion results developed here.

The remaining questions now have a cleaner form: determine optimal relative separation complexity, characterize properness over arbitrary infinite bases, and extend finite-pattern escape and trajectory compression to general finitary and many-sorted contexts. The current formalization establishes the binary compression–escape architecture as a verified starting point for those extensions.

References

- [1] A. Arnold and M. Nivat. The metric space of infinite trees: Algebraic and topological properties. *Fundamenta Informaticae*, 3(4):445–475, 1980. doi: 10.3233/FI-1980-3405.
- [2] J. A. Bergstra and A. Ponse. Division by zero in common meadows, 2014. URL <https://arxiv.org/abs/1406.6878>. Version 4, revised 2021.
- [3] J. A. Bergstra, Y. Hirshfeld, and J. V. Tucker. Meadows and the equational specification of division. *Theoretical Computer Science*, 410(12–13):1261–1271, 2009. doi: 10.1016/j.tcs.2008.12.015.
- [4] W. S. Brainerd. Tree generating regular systems. *Information and Control*, 14(2):217–231, 1969. doi: 10.1016/S0019-9958(69)90065-5.
- [5] A. J. Cain and V. Maltcev. For a few elements more: A survey of finite Rees index, 2013. URL <https://arxiv.org/abs/1307.8259>.
- [6] J. Carlström. Wheels—on division by zero. *Mathematical Structures in Computer Science*, 14(1):143–184, 2004. doi: 10.1017/S0960129503004110.

- [7] L.-T. Chen, J. Adámek, S. Milius, and H. Urbat. Profinite monads, profinite equations, and Reiterman’s theorem. In *Foundations of Software Science and Computation Structures*, volume 9634 of *Lecture Notes in Computer Science*, pages 531–547. Springer, 2016. doi: 10.1007/978-3-662-49630-5_31.
- [8] H. Comon, M. Dauchet, R. Gilleron, F. Jacquemard, D. Lugiez, S. Tison, and M. Tommasi. Tree automata techniques and applications, 2008. URL <https://www.irif.fr/users/tata/>. Electronic monograph.
- [9] B. Courcelle. Fundamental properties of infinite trees. *Theoretical Computer Science*, 25(2): 95–169, 1983. doi: 10.1016/0304-3975(83)90059-2.
- [10] M. Dauchet and S. Tison. The theory of ground rewrite systems is decidable. In *Proceedings of the Fifth Annual IEEE Symposium on Logic in Computer Science*, pages 242–248. IEEE Computer Society Press, 1990. doi: 10.1109/LICS.1990.113750.
- [11] B. A. Davey, M. J. Gouveia, M. Haviar, and H. A. Priestley. Natural extensions and profinite completions of algebras. *Algebra Universalis*, 66(3):205–241, 2011. doi: 10.1007/s00012-011-0155-y.
- [12] R. Diaconescu. An encoding of partial algebras as total algebras. *Information Processing Letters*, 109(23–24):1245–1251, 2009. doi: 10.1016/j.ipl.2009.09.008.
- [13] D. Kozen. Complexity of finitely presented algebras. In *Proceedings of the Ninth Annual ACM Symposium on Theory of Computing*, pages 164–177. ACM, 1977. doi: 10.1145/800105.803406.
- [14] V. Moreau. Profinite trees, through Lawvere theories and the lambda-calculus, 2024. URL <https://arxiv.org/abs/2402.13086>. Version 2, revised 2026.
- [15] N. Ruškuc and R. M. Thomas. Syntactic and Rees indices of subsemigroups. *Journal of Algebra*, 205(2):435–450, 1998. doi: 10.1006/jabr.1997.7392.
- [16] J. C. Vidal and E. C. Llópez. Functoriality of the Schmidt construction. *Logic Journal of the IGPL*, 31(5):822–893, 2023. doi: 10.1093/jigpal/jzac048.